

Gel Electrophoresis Lab

- **Phase 1: Lab orientation**
 1. Identify the lab equipment you will use in this simulation by moving the labels to the correct item
- **Phase 2: Prepare an agarose gel**
 1. Assemble gel casting tray by mounting masking tape in both ends
 2. Add gel comb to create wells for loading DNA samples
 3. Add ethidium bromide to the hot liquid agarose
 4. Carefully pour agarose solution into assembled gel casting tray to prevent formation of air bubbles
 5. Allow gel to solidify for at least 10 minutes.
 6. Remove gel comb and masking tape
 7. Place agarose gel casting tray in electrophoresis chamber
 8. Fill chamber with electrophoresis buffer
- **Phase 3: Load DNA samples**
 1. Place tip onto micro pipette and transfer TNF1 into well 1 of tray. This control allows one to identify which sample is TNF1. Put used tip in Autoclave bag
 2. Place new tip onto micro pipette and transfer TNF2 into well 2 of tray. This control allows one to identify which sample is TNF2. Put used tip in Autoclave bag
 3. Place new tip onto micro pipette and transfer DNA Sample 1 into well 3 of the tray. Put used tip in Autoclave bag
 4. Place new tip onto micro pipette and transfer DNA Sample 2 into well 4 of the tray. Put used tip in Autoclave bag
 5. Place new tip onto micro pipette and transfer DNA Sample 3 into well 5 of the tray. Put used tip in Autoclave bag
- **Phase 4: Separate DNA fragments**
 1. Place lid on electrophoresis chamber
 2. Turn power source on
 3. Set voltage to 70 V by using the arrow keys on the power source
 4. Click play on power source
 5. Run the gel for one hour and thirty minutes until DNA bands have been sufficiently separated to allow identification
 6. Click pause on power source
 7. Turn power source off
 8. Remove lid from electrophoresis chamber
 9. Place gel casting tray onto UV-table
- **Phase 5: Examine the results**
 1. Turn on UV-light using power switch on UV-table

2. Identify genotypes in test samples by matching their DNA bands with DNA bands in TNF1 and TNF2 controls. Record your observations in Lab Data
- **Phase 6: Conclusion**
 1. My Conclusion:
 - **Phase 7: Save Lab Data**
 1. Relevant Lab Data is available to be saved for personal reference. Data will be available if you return to this laboratory simulation

Lab Data	
Sample	Genotype
DNA 1	
DNA 2	
DNA 3	